

AJAY KUMAR S

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Career Profile

Artificial Intelligence and Data Science graduate with hands on experience in developing data driven and AI-based solutions, eager to apply technical knowledge in a dynamic organization while continuously learning and growing professionally.

Education

B.Tech in Artificial Intelligence and Data Science K. Ramakrishnan College of Technology, Trichy - CGPA: 7.52	April 2026
HSC (Bio-Maths) St. Antony's Higher Secondary School, Thanjavur - Percentage: 85.24	May 2021
SSLC (General Subjects) Our Lady of Health Matriculation HSS, Budalur - Percentage: 80.4	March 2019

Skills

Programming Languages: Python, SQL, Java

Frontend Development: HTML, CSS, JavaScript, React.js, Tailwind CSS

Backend Development: Flask, FastAPI, Node.js

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Generative AI, Computer Vision, Reinforcement Learning

Data Science & Analytics: Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Power BI

Database Technologies: MySQL, MongoDB

Tools & Platforms: Git, GitHub, Visual Studio Code, Jupyter Notebook, MySQL Workbench, Postman, Figma

Cloud Services: AWS (EC2, S3)

Soft Skills: Problem Solving, Team Collaboration, Analytical Thinking

Certificates

• Digital Skills Readiness Program - Java Full Stack (TalentNext, Wipro Limited)	View Certificate	Oct 2025
• Introduction to UX/UI Design – IBM	View Certificate	Oct 2025
• Power BI for Beginners: Build Your First Report – Coursera Project Network	View Certificate	Oct 2025
• Applied Generative AI – Simplilearn	View Certificate	May 2025
• AWS Certified Cloud Practitioner	View Certificate	Oct 2024
• DBMS – NPTEL	View Certificate	Mar 2024
• Programming in Python – Jals Network and Technology	View Certificate	Nov 2022
• Anthropic Academy Certifications	View Certificate	Jun 2026
• Microsoft Copilot Foundations – Simplilearn	View Certificate	Nov 2025

Projects

Online Fraud Detection using Random Forest Algorithm 2024

- Developed a machine learning-based fraud detection system using Random Forest to classify legitimate and fraudulent online transactions. Performed data preprocessing, feature engineering, model training, and evaluation on transaction datasets. Designed a real-time fraud prediction workflow to identify suspicious financial activities and reduce false positives.
- Tech Stack:** Python, Scikit-learn, Random Forest, Machine Learning, Data Analysis. [\[GitHub\]](#)

Automated Skin Disease Diagnosis and Treatment Recommendations 2025

- Developed an AI-powered skin disease diagnosis system using MobileNetV2 for image classification and LLaMA 3 via Ollama for disease explanation and treatment recommendations. Implemented image preprocessing, transfer learning, and real-time prediction from dermatological images. Built a privacy-focused solution capable of local inference for healthcare assistance, awareness, and decision support.
- Tech Stack:** Python, TensorFlow, CNN, MobileNetV2, Ollama, LLaMA 3, Computer Vision. [\[GitHub\]](#)

AI-Driven Cybersecurity System to Detect Deepfake and Adversarial Attacks 2025

- Designed a multimodal deepfake detection framework for image, audio, and video analysis using EfficientNet-B4, Transformers, and ensemble learning. Integrated Explainable AI techniques including Grad-CAM, confidence scoring, and watermark-based authenticity verification. Developed a Flask backend with React and Tailwind CSS frontend for real-time media authentication, threat detection, and report generation.
- Tech Stack:** Python, Flask, React, Tailwind CSS, EfficientNet-B4, Transformers, Grad-CAM, Cybersecurity. [\[GitHub\]](#)

Hybrid AI-Based Intrusion Detection and Automated Response System 2026

- Developed a hybrid AI-based intrusion detection system using Autoencoder, XGBoost, and PPO Reinforcement Learning for real-time cyber threat detection and automated response. The system achieved 98.99% detection accuracy while enabling anomaly detection, attack classification, intelligent threat mitigation, and real-time security monitoring through a React and Flask-based dashboard.
- Tech Stack:** Python, Autoencoder, XGBoost, PPO, Reinforcement Learning, TensorFlow, Flask, React, Tailwind CSS, Cybersecurity. [\[GitHub\]](#)

Personal Portfolio Website 2025

- Designed and developed a responsive portfolio website using HTML and CSS to showcase projects, certifications, skills, and achievements across multiple devices.

Achievements

- 1st Runner-Up at Innogen'25 Hackathon (Arczon Industries) Jan 2025
- Certified in Junior Grade English Typewriting

Hobbies

- Meme Creation, Editing